Classical Field Theory — Exercise Sheet 2

Noether currents and their limits

Hints and answers

Use Lecture 2's signs $j^\mu = p_a^\mu Q^a - B^\mu$ and $\partial_\mu j^\mu = -E_a Q^a$. All fields and coefficients below are smooth. The unindexed $Q(t)$ denotes an integrated charge, and Q^a denotes a characteristic. The last problem is a different field theory from Klein–Gordon.

Exercise 2.1 (An improvement can change a boundary charge). Let j^μ be a current in $1+d$ dimensions and let $K^{\mu\nu} = -K^{\nu\mu}$ be smooth. Define $\widetilde{j}^\mu = j^\mu + \partial_\nu K^{\mu\nu}$. An added current with identically zero divergence is called an *identically conserved improvement* in this problem.

(a) Prove that the two currents have the same divergence for every field. Show that their charges agree on the spatial torus when K is periodic. On $\mathbb{R}^d$, assume the original slice charge exists and show equality when $K^{0i}(t,\cdot)$ is compactly supported.

(b) In $1+1$ dimensions take $K^{01} = t/(1+x^2)$ and $K^{10} = -K^{01}$. Compute the added density and flux. Find the change in total charge on $\mathbb{R}$ and the change in charge on the fixed interval $[0,1]$. Does an identically conserved improvement leave every regional charge unchanged?

Hint. Pair the terms with interchanged indices in the double divergence. The density change is a spatial derivative. Integrate that derivative before taking any limit at infinity.

Answer. The divergence change is zero. The added components in the example are $-2tx/(1+x^2)^2$ and $-1/(1+x^2)$. The total charge change is zero; on $[0,1]$ it is $-t/2$.

Exercise 2.2 (The converse with an off-shell identity supplied). For a first-order density, suppose a legitimate characteristic Q^a and specified local current j^μ satisfy $\partial_\mu j^\mu = -E_a Q^a$ for arbitrary fields, before imposing their equations. Prove that $B^\mu = p_a^\mu Q^a - j^\mu$ gives $\delta_Q L = \partial_\mu B^\mu$.

Apply this calculation on $\mathbb{R}^{1,1}$ to $L = \frac{1}{2}(\phi_t^2 - \phi_x^2)$, $\eta(t,x) = t + x$, and

$$j^0 = \eta\phi_t - \phi, \qquad j^1 = -\eta\phi_x + \phi.$$

Find its characteristic, the divergence term and an actual one-parameter point transformation realising that characteristic. Verify the off-shell identity directly.

For context, the following broader correspondence is a *cited input*, not a task to prove: in the local finite-derivative formulation for the scalar wave Euler–Lagrange equation, conservation-law classes have generalized variational characteristics. Currents are identified modulo currents vanishing on solutions and identically conserved improvements; characteristics are identified modulo ones vanishing on solutions. Here local means depending smoothly on the coordinates, field and finitely many of its derivatives. This is the scalar wave case of Olver, §5.3, Theorem 5.58, p. 334. Explain why the assertion that an arbitrary current is conserved on solutions is not the supplied off-shell identity used in the first calculation.

Hint. Start with the pointwise first-variation identity of Lecture 1. For the example, calculate the divergence before using $\phi_{tt} = \phi_{xx}$. A field shift by a prescribed function can depend on the spacetime point.

Answer. The example has $Q^1 = \eta$, $B^0 = \phi$ and $B^1 = -\phi$. The transformation is $\phi_\varepsilon(t,x) = \phi(t,x) + \varepsilon(t+x)$. Its current has divergence $\eta(\phi_{tt} - \phi_{xx}) = -E\eta$.

Exercise 2.3 (A Noether charge escapes to infinity). Let p and q be real fields on $\mathbb{R}_t \times \mathbb{R}_x$, with density $L = p(q_t + x^2 q_x)$. Choose $p_0 \in C_c^\infty((1,2))$, nonnegative and with integral one, and extend it by zero to $\mathbb{R}$. Set $q = 0$ and

$$p(t,x) = \begin{cases} (1+tx)^{-2} p_0\big(x/(1+tx)\big), & 1+tx > 0, \\ 0, & 1+tx \le 0. \end{cases}$$

(a) Derive the two field equations. Find the Noether current for the constant shift of q. Prove that the displayed fields are smooth at every finite spacetime point and satisfy both equations.

(b) Compute $Q(t) = \int_{\mathbb{R}} p(t,x)\mathrm{d}x$. Find its derivative for $t > 0$ and the limiting flux as $x \to +\infty$. Show that the initial charge is one but the charge is zero for $t \ge 1$.

(c) Identify exactly which full-space charge hypothesis in Lecture 2 fails. Why is this neither an off-shell example nor a claim about Klein–Gordon propagation?

Hint. For the transport equation use $x = \xi/(1 - t\xi)$ and retain its Jacobian. Near a point where $1 + tx = 0$, the profile argument has large absolute value, so the compact support of p_0 removes the apparent singularity. Do not differentiate the full-space integral before computing it.

Answer. The equations are $q_t + x^2 q_x = 0$ and $p_t + \partial_x(x^2 p) = 0$. The shift current is $(p, x^2 p)$. For $t \le 0$, $Q = 1$; for $t > 0$, $Q = \int_0^{1/t} p_0(\xi)\mathrm{d}\xi$. Its derivative is $-p_0(1/t)/t^2$, the negative outgoing flux. Where that flux is positive, the spatial current is not in L^1.